

Work done by force = $Fs = 3.0 \times 0.5 = 1.5 \text{ J}$

- 10 C** The resultant of 160 N and 120 N should provide the horizontal centripetal force.

Resultant force = $\sqrt{160^2 - 120^2} = 105.8 \text{ N}$

Resultant force = ma

$105.8 = (120/9.81) a \Rightarrow a = 8.65 \text{ ms}^{-2}$

- 11 A** Gravitational potential increases towards infinity. -20 MJ kg^{-1} is at a higher point than -30

MJ kg^{-1} . So mass is moving away from Earth